From Abstracts to Rooms: Poster Session Assignment by Semantic Similarity and Optimisation

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1 INTRODUCTION

A large poster session may hold thousands of papers across rooms of different sizes. Similar topics belong close together so visitors can find related work, but building that layout by hand is impractical at this scale.

Author-chosen subject labels cannot solve it either, since the areas are wildly uneven. The largest area in the data used here holds **1,685** papers, the smallest only **54**, so any room plan must divide large areas and combine small ones.

CVPR already groups its posters with paper embeddings [1], yet reports no numerical evaluation of the result. **Nobody measures whether the rooms came out coherent.** Whether a program can take any conference's papers and any set of rooms and return a plan in which every room holds papers about similar things is the question this work answers.

2 OBJECTIVE

Can a program take any conference's papers and any set of room sizes and return a seating plan in which every room holds papers about similar things?

The delivered plan must be:

1. **Coherent** — papers sharing a room talk about similar things.
2. **Feasible** — every paper seated, no room over capacity, no room empty.
3. **Fast** — holds at the scale of a real conference, thousands of papers.
4. **General** — any conference's papers, any set of room sizes, no retuning.
5. **Reproducible** — repeated runs follow the same search.

3 METHODS

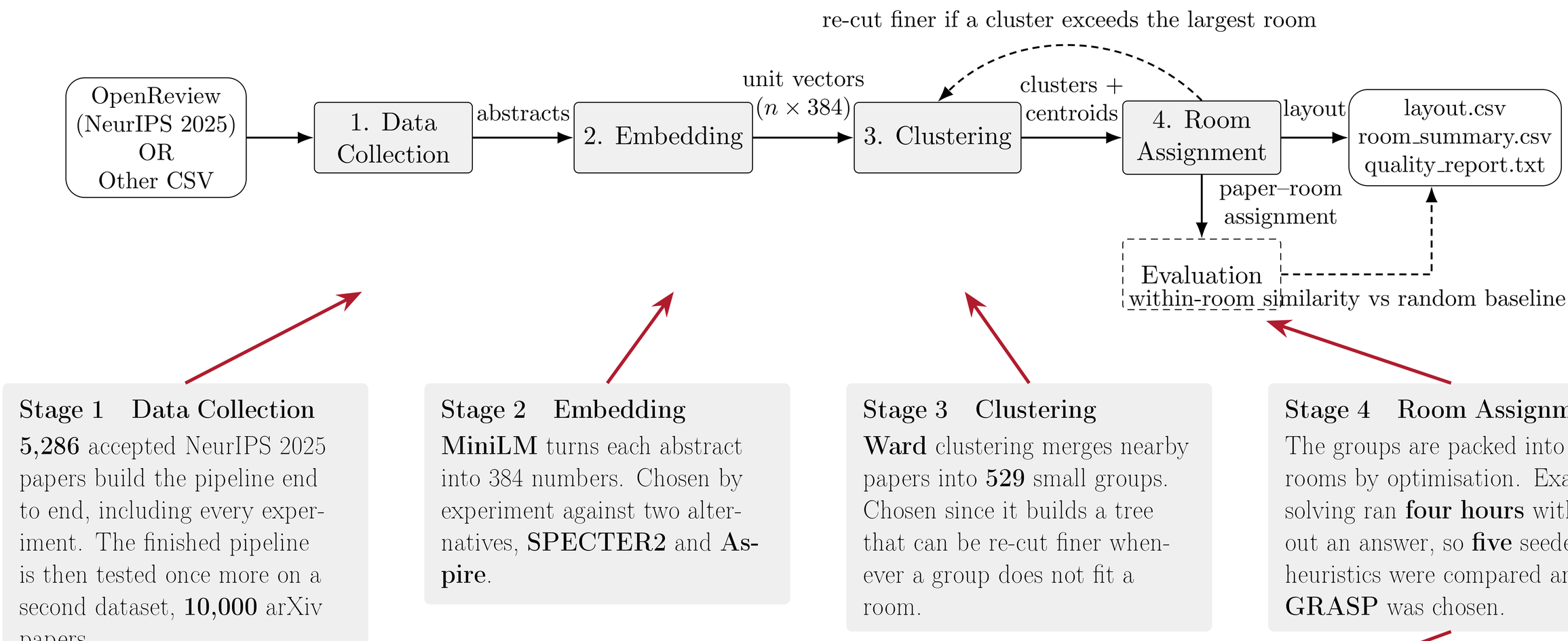


Figure 1. All stages of the dissertation pipeline.

What is maximised. Each group is summarised by its centroid, the average of its papers' vectors. The similarity w_{ij} of two groups is the cosine similarity of their centroids, the comparison these embeddings are built for [2]. Every pair of groups that shares a room adds its similarity to one total, and that total is made as large as possible.

$$f(a) = \sum_{\substack{1 \le i < j \le C \\ a_i = a_j}} W_{ij}$$

subject to: no room over its seats, and no room left empty.

4 RESULTS

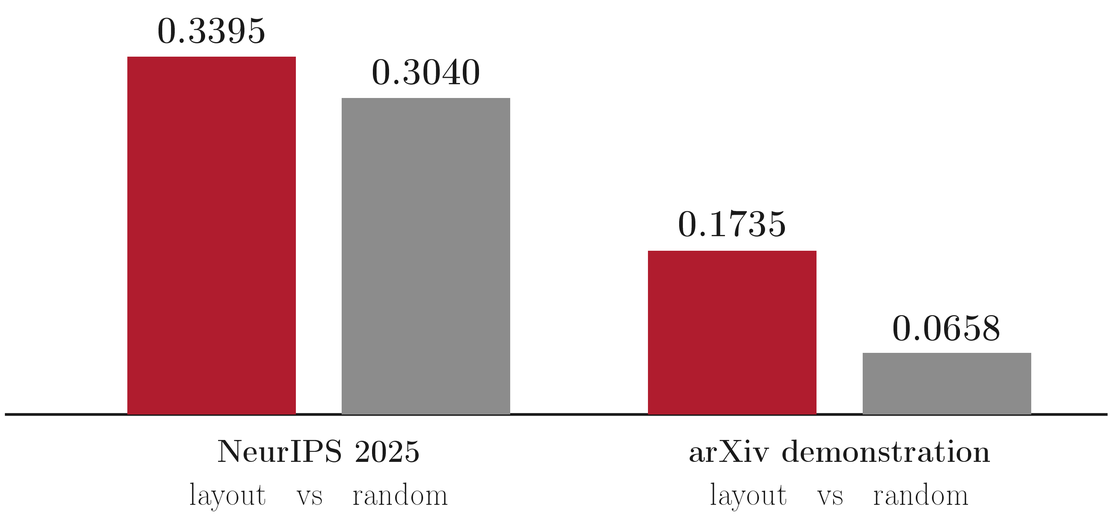


Figure 2. Within-room paper similarity against a random assignment of the same papers into the same rooms.

5 CONCLUSIONS

The research question is answered. A program took two different sets of papers and two different sets of rooms, and returned plans in which the rooms hold papers about similar things.

- ✓ **Coherent** — both layouts score above their random baseline.
- ✓ **Feasible** — every paper seated, no room over capacity, none empty.
- ✓ **Fast** — minutes on a laptop, not four hours without an answer.
- ✓ **General** — the second dataset needed no change to the tool.
- ✓ **Reproducible** — repeated runs follow the same search.

References. [1] CVPR, poster session grouping, 2024. [2] Reimers & Gurevych, *Sentence-BERT*, EMNLP 2019. [3] Friedman, *JASA*, 1937. [4] Holm, *Scand. J. Statist.*, 1979.

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Code. github.com/sejdinii/poster-session